

Tutorial – 2

1. Predict and write output for the following code.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q1
    {
        static void Main(string[] args)
        {
            /* local variable definition */
            int a = 10;

            /* check the boolean condition using if statement */
            if (a < 20)
            {
                /* if condition is true then print the following */
                Console.WriteLine("a is less than 20");
            }
            Console.WriteLine("value of a is : {0}", a);
            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>csc Q1.cs
Microsoft (R) Visual C# Compiler version 4.14.0-3.25279.5 (995f12b6)
Copyright (C) Microsoft Corporation. All rights reserved.

D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q1
a is less than 20
value of a is : 10
```

2. Write missing statement to get the desired output.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q2
    {
        static void Main(string[] args)
```

```
{
    /* local variable definition */
    int a = 100;

    /* check the boolean condition */
    if (a < 20)
    {
        /* if condition is true then print the following */
        Console.WriteLine("a is less than 20");
    }
    else
    {
        /* if condition is false then print the following */
        Console.WriteLine("a is not less than 20"); // Missing statement-1
    }
    Console.WriteLine("value of a is : {0}", a); // Missing statement-2
    Console.ReadLine();
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q1
a is less than 20
value of a is : 10
```

3. Correct the following code and write output for the corrected code.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q3
    {
        static void Main(string[] args)
        {
            string firstName = "John";
            string lastName = "Doe";
            Console.WriteLine("Name: " + firstName + " " + lastName);
            Console.WriteLine("Please enter a new first name:");
            firstName = Console.ReadLine();
            Console.WriteLine("New name: " + firstName + " " + lastName);
            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q3
Name: John Doe
Please enter a new first name:
Saurav
New name: Saurav Doe
```

4. Input two number A and B. perform different operations using different operators and different data types available in C#. (Note : Follow all the operators and data types to do above task. Use Online help whenever necessary.)

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q4
    {
        static void Main(string[] args)
        {
            Console.Write("Enter number A: ");
            int A = Convert.ToInt32(Console.ReadLine());

            Console.Write("Enter number B: ");
            int B = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("\n=== Arithmetic Operators ===");
            Console.WriteLine("A + B = " + (A + B));
            Console.WriteLine("A - B = " + (A - B));
            Console.WriteLine("A * B = " + (A * B));

            Console.WriteLine("A / B = " + ((double)A / B));

            Console.WriteLine("\n=== Relational Operators ===");
            Console.WriteLine("A > B : " + (A > B));
            Console.WriteLine("A == B : " + (A == B));

            Console.WriteLine("\n=== Logical Operator ===");
            Console.WriteLine("A>0 && B>0 : " + (A > 0 && B > 0));

            Console.WriteLine("\n=== Bitwise Operator ===");
            Console.WriteLine("A & B = " + (A & B));

            Console.WriteLine("\n=== Other Data Types ===");
            double d = (double)A / B;
```

24SOECE13901

Enterprise Computing Through .NET Framework (CE525)

```
Console.WriteLine("Double division: " + d);
string str = "Numbers: " + A + " and " + B;
Console.WriteLine("String example: " + str);

bool flag = (A == B);
Console.WriteLine("Boolean example (A==B): " + flag);
Console.ReadLine();
    }
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q4
Enter number A: 10
Enter number B: 5

=== Arithmetic Operators ===
A + B = 15
A - B = 5
A * B = 50
A / B = 2

=== Relational Operators ===
A > B : True
A == B : False

=== Logical Operator ===
A>0 && B>0 : True

=== Bitwise Operator ===
A & B = 0

=== Other Data Types ===
Double division: 2
String example: Numbers: 10 and 5
Boolean example (A==B): False
```

5. Rearrange the given code to correct the program. The resultant program will be to enter 5 elements into an array and print sum of these elements.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
```

```
{
    internal class Q5
```

```
{
    static void Main(string[] args)
    {
        int[] arr = new int[5];
        int sum = 0;

        for (int i = 0; i < 5; i++)
        {
            Console.Write("Enter Element {0}: ", i + 1);
            string str = Console.ReadLine();
            arr[i] = Convert.ToInt32(str);
        }

        for (int i = 0; i < 5; i++)
        {
            sum = sum + arr[i];
        }

        Console.WriteLine("Sum of Elements : {0}", sum);
        Console.ReadLine();
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q5
Enter Element 1: 10
Enter Element 2: 20
Enter Element 3: 30
Enter Element 4: 40
Enter Element 5: 50
Sum of Elements : 150
```

6. Write missing statement to get the desired output.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q6
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Hello, World!");
        }
    }
}
```

```
Console.WriteLine("You entered the following {0} command line arguments:",args.Length);
```

```
    for (int i = 0; i < args.Length; i++)          // Missing statement-1
    {                                              // Missing statement-2
        Console.WriteLine(args[i]);              // Missing statement-3
    }                                              // Missing statement-4
}
}
```

Output:

```
D:\RKU\5th SEM\.NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q6 Saurav Pandit
Hello, World!
You entered the following 2 command line arguments:
Saurav
Pandit
```

7. Predict and write the output of the given code.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q7
    {
        public void swap(ref int x, ref int y)
        {
            int temp;
            temp = x; /* save the value of x */
            x = y; /* put y into x */
            y = temp; /* put temp into y */
        }
    }
}
class TestRef
{
    static void Main(string[] args)
    {
        Q7 n = new Q7();
        /* local variable definition */
        int a = 100;
        int b = 200;

        Console.WriteLine("Before swap, value of a : {0}", a);
        Console.WriteLine("Before swap, value of b : {0}", b);
    }
}
```

24SOECE13901

Enterprise Computing Through .NET Framework (CE525)

```
/* calling a function to swap the values */  
  
n.swap(ref a, ref b);  
Console.WriteLine("After swap, value of a : {0}", a);  
Console.WriteLine("After swap, value of b : {0}", b);  
Console.ReadLine();  
}  
}  
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q7  
Before swap, value of a : 100  
Before swap, value of b : 200  
After swap, value of a : 200  
After swap, value of b : 100
```

8. Find out error code and correct it. Write the output of the corrected code.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2  
{  
    internal class Q8  
    {  
        public int getValues(out int x, out int y, out int z)  
        {  
            Console.WriteLine("Enter the first value: ");  
            x = Convert.ToInt32(Console.ReadLine());  
  
            Console.WriteLine("Enter the second value: ");  
            y = Convert.ToInt32(Console.ReadLine());  
  
            Console.WriteLine("Enter the third value: ");  
            z = Convert.ToInt32(Console.ReadLine());  
  
            int sum = x + y + z;  
            return sum;  
        }  
    }  
}  
  
class TestOut  
{  
    static void Main(string[] args)
```

```
{
    Q8 n = new Q8();

    int a, b, c, sum;
    sum = n.getValues(out a, out b, out c);
    Console.WriteLine("After method call, value of a : {0}", a);
    Console.WriteLine("After method call, value of b : {0}", b);
    Console.WriteLine("After method call, value of c : {0}", c);
    Console.WriteLine("Sum : {0}", sum);
    Console.ReadLine();
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q8
Enter the first value:
100
Enter the second value:
200
Enter the third value:
300
After method call, value of a : 100
After method call, value of b : 200
After method call, value of c : 300
Sum : 600
```

9. Given an array A containing $2*N+2$ positive numbers, out of which $2*N$ numbers exist in pairs whereas the other two number occur exactly once and are distinct. Find the other two numbers.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q9
    {
        static void UniqueNumber(int[] arr)
        {
            int xor = 0;
            foreach (int num in arr)
                xor ^= num;

            // Find rightmost set bit
```


24SOECE13901

Enterprise Computing Through .NET Framework (CE525)

```
int rightmostSetBit = xor & -xor;
```

```
int num1 = 0, num2 = 0;
```

```
foreach (int num in arr)
{
    if ((num & rightmostSetBit) != 0)
        num1 ^= num;
    else
        num2 ^= num;
}
```

```
Console.WriteLine(num1 + " " + num2);
}
```

```
static void Main(string[] args)
{
    int[] arr1 = { 1, 2, 3, 2, 1, 4 };
    Console.Write("Output for Example 1: ");
    UniqueNumber(arr1);
```

```
    int[] arr2 = { 2, 1, 3, 2 };
    Console.Write("Output for Example 2: ");
    UniqueNumber(arr2);
```

```
    Console.ReadLine();
}
```

```
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q9
Output for Example 1: 3 4
Output for Example 2: 3 1
```

10. Given a matrix `mat[][]` of size `N x M`, where every row and column is sorted in increasing order, and a number `X` is given. The task is to find whether element `X` is present in the matrix or not.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
```

```
{
    internal class Q10
    {
        static bool SearchMatrix(int[,] mat, int n, int m, int x)

        {
            int row = 0;    // start from top-right corner
            int col = m - 1;

            while (row < n && col >= 0)
            {
                if (mat[row, col] == x)
                    return true;
                else if (mat[row, col] > x)
                    col--;
                else
                    row++;
            }
            return false;
        }
    }

    static void Main()
    {
        int[,] mat = {
            { 10, 20, 30, 40 },
            { 15, 25, 35, 45 },
            { 27, 29, 37, 48 },
            { 32, 33, 39, 50 }
        };

        int n = mat.GetLength(0);
        int m = mat.GetLength(1);

        Console.WriteLine("Enter element to search: ");
        int x = int.Parse(Console.ReadLine());

        if (SearchMatrix(mat, n, m, x))
            Console.WriteLine("Element {0} is present in matrix.", x);
        else
            Console.WriteLine("Element {0} is NOT present in matrix.", x);
    }
}
```

```
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q10
Enter element to search: 35
Element 35 is present in matrix.

D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>csc Q10.cs
Microsoft (R) Visual C# Compiler version 4.14.0-3.25279.5 (995f12b6)
Copyright (C) Microsoft Corporation. All rights reserved.

D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q10
Enter element to search: 100
Element 100 is NOT present in matrix.
```

11. Write a program to find the sum of N elements of an Array.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q11
    {
        static void Main(string[] args)
        {
            Console.Write("Enter number of elements: ");
            int n = Convert.ToInt32(Console.ReadLine());

            int[] arr = new int[n];
            int sum = 0;

            for (int i = 0; i < n; i++)
            {
                Console.Write("Enter element {0}: ", i + 1);
                arr[i] = Convert.ToInt32(Console.ReadLine());
                sum += arr[i];
            }

            Console.WriteLine("Sum of array elements = " + sum);

            Console.ReadLine();
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q11
Enter number of elements: 4
Enter element 1: 10
Enter element 2: 20
Enter element 3: 30
Enter element 4: 40
Sum of array elements = 100
```

12. Write a program to find the element from an Array and print 1 if element is found else print 0.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q12
    {
        static void Main()
        {
            Console.Write("Enter size of array: ");
            int n = int.Parse(Console.ReadLine());

            int[] arr = new int[n];

            Console.WriteLine("Enter array elements:");
            for (int i = 0; i < n; i++)
            {
                arr[i] = int.Parse(Console.ReadLine());
            }

            Console.Write("Enter element to search: ");
            int x = int.Parse(Console.ReadLine());

            bool found = false;

            // Linear Search
            for (int i = 0; i < n; i++)
            {
                if (arr[i] == x)
                {
                    found = true;
                }
            }
        }
    }
}
```

24SOECE13901

Enterprise Computing Through .NET Framework (CE525)

```
        break;
    }
}

if (found)
    Console.WriteLine("1"); // Element found
else
    Console.WriteLine("0"); // Element not found
}
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q12
Enter size of array: 4
Enter array elements:
11
21
31
41
Enter element to search: 80
0
```

- 13. Write a Program that will accept the amount and find how many minimum no of notes you required for that.**

(Using the rupee notes of 1,2,5,10,20,50,100,500,2000)

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q13
    {
        static void Main()
        {
            Console.Write("Enter the amount: ");
            int amount = int.Parse(Console.ReadLine());

            int[] notes = { 2000, 500, 200, 100, 50, 20, 10, 5, 2, 1 };

            foreach (int note in notes)
            {
                int count = amount / note; // how many notes of this type
                amount = amount % note;    // remaining amount
            }
        }
    }
}
```

```
        Console.WriteLine("Notes of Rs.{0} = {1}", note, count);
    }
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q13
Enter the amount: 578
Notes of Rs.2000 = 0
Notes of Rs.500 = 1
Notes of Rs.200 = 0
Notes of Rs.100 = 0
Notes of Rs.50 = 1
Notes of Rs.20 = 1
Notes of Rs.10 = 0
Notes of Rs.5 = 1
Notes of Rs.2 = 1
Notes of Rs.1 = 1
```

14. Write a Program to find the eligibility of admission for a professional course based on the following criteria:

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q14
    {
        static void Main()
        {
            Console.Write("Input the marks obtained in Maths: ");
            int maths = int.Parse(Console.ReadLine());

            Console.Write("Input the marks obtained in Physics: ");
            int physics = int.Parse(Console.ReadLine());

            Console.Write("Input the marks obtained in Chemistry: ");
            int chemistry = int.Parse(Console.ReadLine());

            int total = maths + physics + chemistry;
            int mpTotal = maths + physics;
```

24SOECE13901

Enterprise Computing Through .NET Framework (CE525)

```
if (maths >= 65 && physics >= 55 && chemistry >= 50 &&
    (total >= 180 || mpTotal >= 140))
{
    Console.WriteLine("The candidate is eligible for admission.");
}
else
{
    Console.WriteLine("The candidate is not eligible for admission.");
}
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q14
Input the marks obtained in Maths: 85
Input the marks obtained in Physics: 57
Input the marks obtained in Chemistry: 63
The candidate is eligible for admission.
```

15. Write a Program which accepts name from the user and prints the same.

```
namespace _24SOECE13901_SAURAV_PANDIT_TUTORIAL_2
{
    internal class Q15
    {
        static void Main()
        {
            Console.Write("Enter your name: ");
            string name = Console.ReadLine();

            Console.WriteLine("You entered: " + name);
        }
    }
}
```

Output:

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_TUTORIAL_2>Q15
Enter your name: Saurav Pandit
You entered: Saurav Pandit
```